Professional Installation Guide: Enterprise Server Deployment & Configuration

1. Project Overview

This comprehensive installation guide documents the design, deployment, verification, and comparative analysis of enterprise server operating systems. The project establishes foundational competencies in virtualization using Oracle VM VirtualBox, contrasts traditional BIOS with modern UEFI firmware, maps the low-level Linux boot architecture, and deploys both Ubuntu Server and Windows Server Evaluation environments to evaluate real-world enterprise architectures.

2. Objectives

Configure and provision virtualized enterprise server instances (server01 and WINSERVER01) adhering to strict resource allocations.

1. -Configure and provision virtualized enterprise server instances (`server01` and
2. `WINSERVER01`) adhering to strict resource allocations.
3. -Install and verify Ubuntu Server 24.04 LTS, confirming hostnames, static/dynamic IP

networking, OpenSSH connectivity, and package updates.

4. -Contrast traditional BIOS and modern UEFI architectures across performance, security, and

storage capacity metrics.

5. -Illustrate the 8-stage sequential Linux boot process through a standardized technical

flowchart.

6. -Deploy Windows Server Evaluation, resolving partition allocation errors and validating
7. configuration parameters via Server Core (`SConfig`).
8. -Perform a deep-dive evaluation of Windows Server, Ubuntu Server, and Rocky Linux across
9. licensing, user interfaces, security, performance, and enterprise use cases.

2. Software Requirements

1. -Hypervisor: Oracle VM VirtualBox (latest stable release)
2. -Guest Operating System 1: Ubuntu Server (Long Term Support ISO)
3. -Guest Operating System 2: Microsoft Windows Server Evaluation ISO

(e.g., Windows Server 2025 Standard/Datacenter Evaluation)

4. -Diagramming Tool: Draw.io / Lucidchart (for boot process flowcharts)
5. -Documentation Tools: Markdown editor, Microsoft Word / PDF export

Utility

4. Virtual Machine Specifications

Specification	Ubuntu Server (`server01`)	Windows Server (`WINSERVER01`)
Hypervisor	Oracle VM VirtualBox	Oracle VM VirtualBox
RAM Allocation	4096 MB (4 GB)	4096 MB (4 GB)
Processor Cores (CPU)	2 Virtual CPUs	2 Virtual CPUs
Virtual Hard Disk Storage	40 GB (Dynamic/Fixed)	50 GB (Dynamic/Fixed)
Network Configuration	NAT (`enp0s3`)	NAT Adapter
Firmware Standard	UEFI / BIOS Compatible	UEFI

5. Step-by-Step Installation Procedure

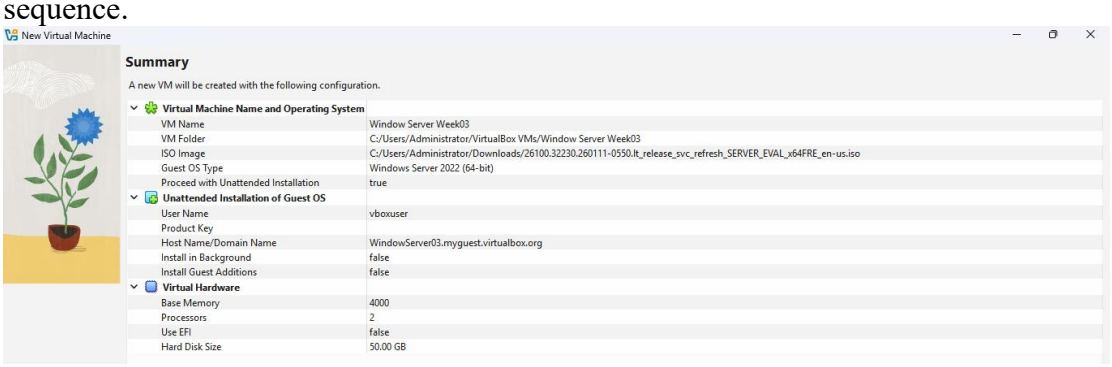
Part A: Ubuntu Server Installation

1. **VM Creation:** Open VirtualBox, click *New*, assign the name `server01`, allocate 4GB RAM, 2 CPUs, and attach the Ubuntu Server ISO image.
2. **Storage Allocation:** Set the virtual disk size to 40 GB.
3. **OS Setup Wizard:** Boot the VM, select language, keyboard layout, and choose the standard Ubuntu Server installation profile.
4. **Network & Storage Config:** Accept default DHCP networking and configure the primary disk layout.
5. **User Profile Creation:** Set the local administrative username (`vboxuser`) and assign a secure master password.
6. **OpenSSH Server:** Ensure the prompt to install the OpenSSH server package is checked during setup.
7. **Finalization:** Complete the installation and reboot the virtual machine.



Part B: Windows Server Installation

1. **VM Creation:** Create a new VirtualBox machine named `WINSERVER01`, allocate 4GB RAM, 2 CPUs, and attach the Windows Server Evaluation ISO.
2. **Storage Allocation:** Provision a 50 GB virtual hard disk.
3. **Setup Execution:** Boot the VM, select language and keyboard preferences, and click *Install Now*.
4. **Edition Selection:** Choose **Windows Server Standard/Datacenter Evaluation (Desktop Experience)** or Server Core based on administrative preference.
5. **Partition Management:** Resolve any formatting errors by deleting legacy partitions until the drive reads as unallocated space.
6. **Password & Login:** Assign a strong administrator password and complete the first-time login
   sequence.



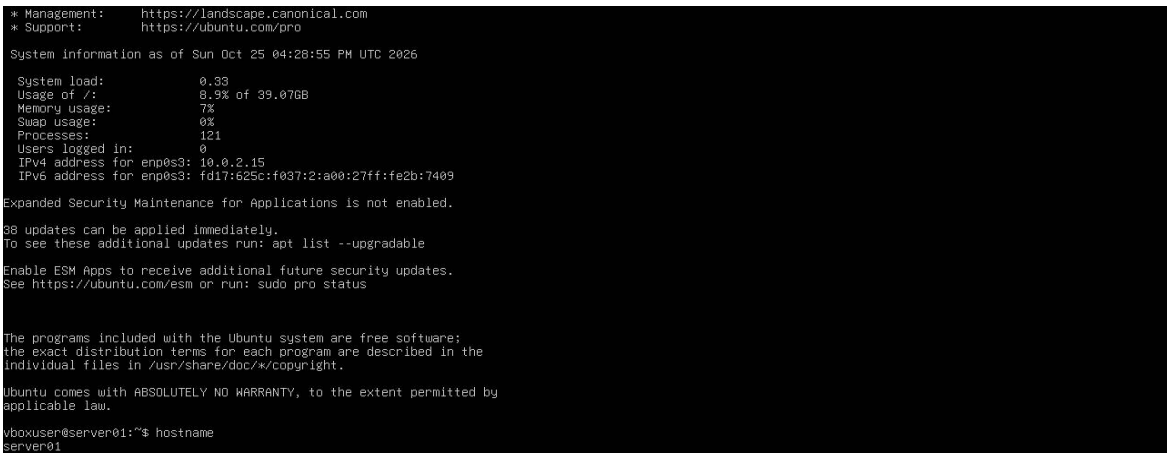
6. Configuration Details

1. **Ubuntu Server Hostname:** Configured explicitly to `server01`.
2. **Windows Server Computer Name:** Configured to `WINSERVER01`.
3. **Network Interfaces:** Utilizing NAT adapter configurations (`enp0s3` on Ubuntu yielding dynamic IP addresses such as `10.0.2.15`).
4. **Remote Access:** OpenSSH daemon enabled and running on Ubuntu Server for secure terminal administration.

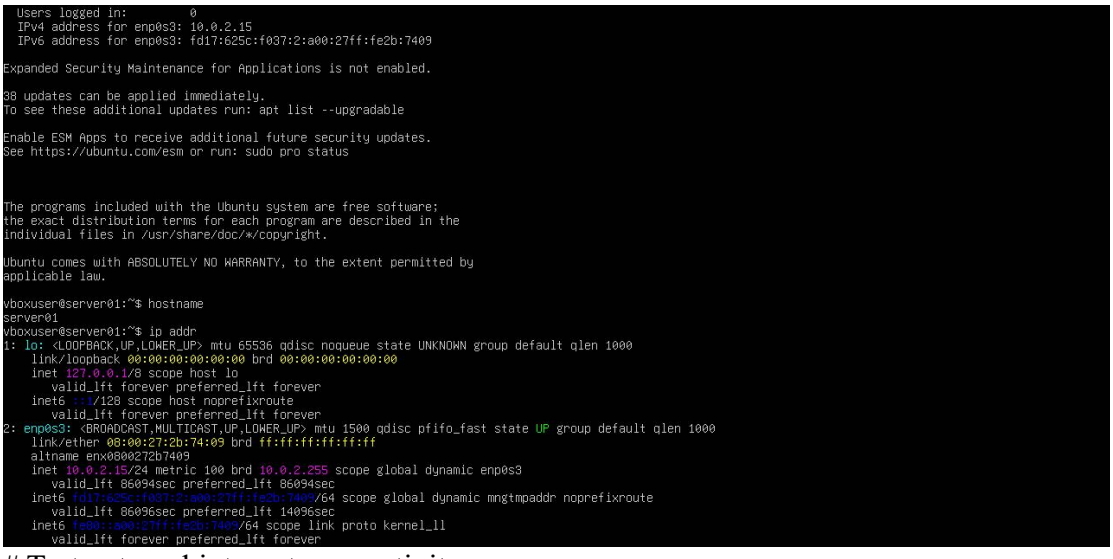
7. Verification Commands

Run the following verification sequence on Ubuntu Server to validate system readiness:

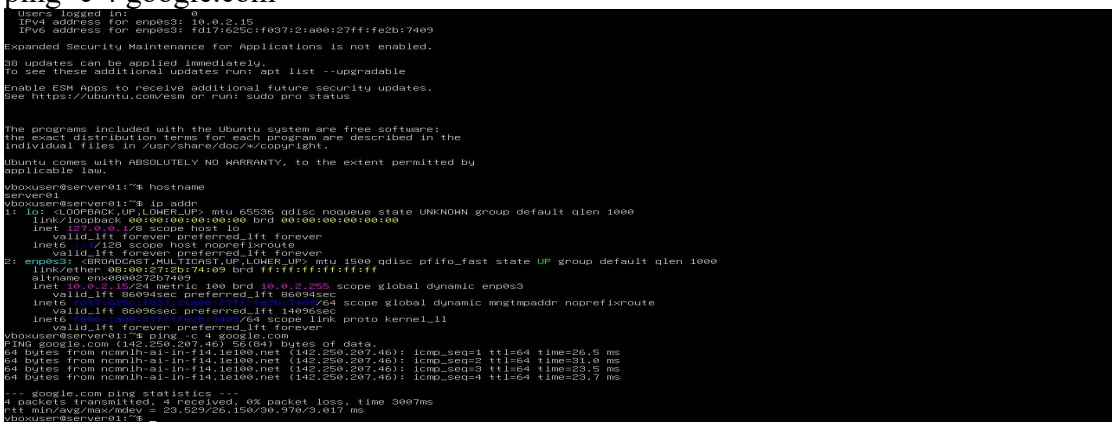
Verify hostname assignment
hostname



Verify IP addressing and active network interfaces
ip addr

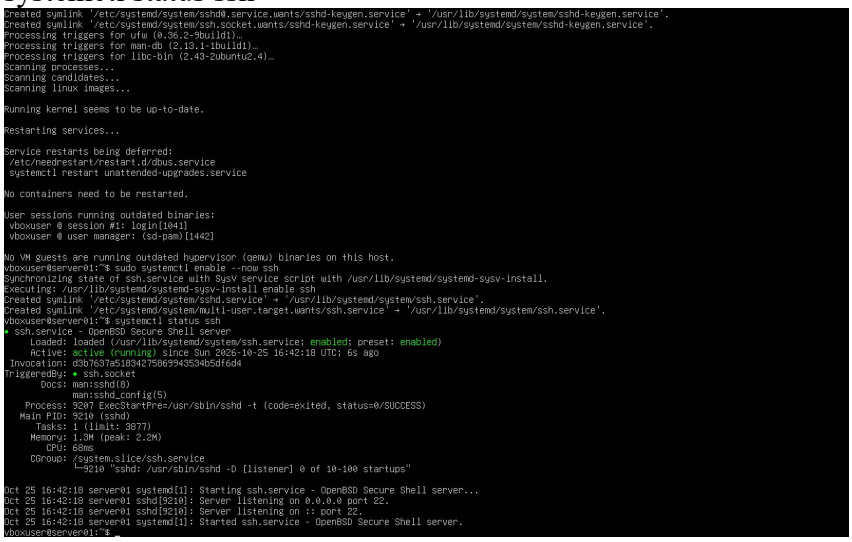


Test external internet connectivity
ping -c 4 google.com



Update package repositories and upgrade software
sudo apt update
sudo apt upgrade -y

Verify OpenSSH service status
systemctl status ssh



8. BIOS vs. UEFI Comparison

Feature / Criteria	Traditional BIOS	Modern UEFI
Definition	Legacy 16-bit firmware interface stored on a ROM chip.	Modern 32/64-bit extensible firmware standard.
Boot Process	Sequential hardware polling; reads MBR boot sector.	Parallel initialization; executes `.efi` binaries from ESP.
Maximum Disk Support	2.2 Terabytes (TB) limitation.	9+ Zettabytes capacity.
Partition Style	MBR (Master Boot Record).	GPT (GUID Partition Table, up to 128 partitions).
Security Features	Basic passwords; no pre-boot integrity validation.	Secure Boot with cryptographic signature verification.
Speed	Slower initialization due to legacy constraints.	Significantly faster startup times.
Modern Usage	Obsolete; emulated via CSM for legacy systems.	Universal industry standard for modern hardware.

9. Boot Process Flowchart

The 8 sequential stages of the Ubuntu Server boot process are mapped below:

1. **Power On:** Motherboard power supply and CPU reset vector execution.
2. **BIOS/UEFI Initialization:** POST (Power-On Self-Test) and hardware bus verification.
3. **Boot Device Detection:** Scanning storage media (MBR sector or ESP partition).
4. **Boot Loader (GRUB):** GRUB2 menu loads kernel (`vmlinuz`) and `initramfs` into RAM.
5. **Linux Kernel:** Kernel uncompresses, takes hardware control, and mounts the root file system (`/`).
6. **init / systemd (PID 1):** `systemd` initializes as Process ID 1 to manage user-space operations.
7. **Services Start:** Mounting file systems, configuring networks, and starting daemon processes (OpenSSH).
8. **Login Prompt:** Initializing virtual terminal getty services and displaying `server01 login:`.



10. Windows Server Installation Summary

The Windows Server Evaluation deployment was successfully executed on VirtualBox with 4GB RAM, 2 CPUs, and a 50GB virtual disk. Following partition cleanup via installer tools or `diskpart`, the operating system was deployed, establishing the computer name `WINSERVER01`, enforcing administrative credentials, and validating system manageability via Server Core (`SConfig`).

11. OS Comparison Table

Criteria	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Commercial / Proprietary (Paid CALs/Cores).	Open Source (GPL, Free).	Open Source (GPL, Free RHEL clone).
User Interface	GUI / PowerShell / Windows Admin Center.	Headless CLI (Terminal).	Headless CLI (Bash / Shell).
Package Manager	MSI, Windows Update, `winget`.	APT (`apt`, `.deb`).	DNF / YUM (`.rpm`).
Security	Active Directory, Windows Defender, BitLocker.	UFW, AppArmor, SSH keys.	SELinux, firewalld, enterprise auditing.
Use Cases	Active Directory, IIS, SQL Server, Windows apps.	Cloud hosting, Docker/Kubernetes, microservices.	Database clusters, HPC, enterprise backends.

12. Problems Encountered

1. **Missing OpenSSH Package:** Initial verification on Ubuntu Server returned a `Unit ssh.service could not be found` error because the package was omitted during custom installation prompts.
2. **Partition Allocation Error:** During Windows Server installation, VirtualBox threw an error stating, *"There is an error selecting this partition for install. Please select a different partition on a refresh."*

13. Solutions Implemented

1. **SSH Resolution:** Updated local package lists and installed the OpenSSH server package explicitly via terminal commands (`sudo apt update \&\& sudo apt install openssh-server -y`), followed by enabling the service (`systemd`).
2. **Partition Resolution:** Cleared residual table metadata inside the Windows installer by deleting existing legacy partitions until the drive displayed entirely as unallocated space, then refreshed the volume view.

14. Lessons Learned

1. Virtualization resource planning requires careful balancing of hypervisor host limits against guest requirements (allocating heavier RAM/storage footprints to Windows Server than minimal headless Linux instances).
2. Low-level firmware architecture differences (BIOS vs. UEFI) directly dictate storage scalability and boot security postures.
3. Troubleshooting system service errors reinforces the importance of package manager proficiency and command-line diagnostics in enterprise administration.

15. References

6. Canonical Ltd. (2026). *Ubuntu Server Documentation & Installation Guides*. Retrieved from official Ubuntu documentation repositories.
7. Microsoft Corporation (2026). *Windows Server Evaluation Center & Deployment Guidelines*. Retrieved from Microsoft Azure / Documentation portals.
8. Oracle Corporation (2026). *Oracle VM VirtualBox User Manual*. Virtualization configuration guidelines.